Pujan Ajmera

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Education

University of California, Los Angeles (UCLA) 2022 – Expected 2027

Ph.D. Chemistry, Theory & Computation, GPA: 4.00/4.00

· Advisor: Anastassia N. Alexandrova

University of Michigan, Ann Arbor (UMich) 2018 – 2021

B.S.E. Engineering Physics, Summa Cum Laude

· Minor: Biophysics

Research Experience

Graduate Researcher, University of California, Los Angeles 2022 – Present

Alexandrova Lab

· Understanding enzyme control of electronic structure in natural and artificial metalloenzymes using multiconfigurational approaches

· Developing theory and methods of analyzing dynamics of 3D heterogeneous, electric fields in enzyme active sites and their impact on chemical reactivity, notably charge transfer

Research Assistant, University of Michigan, Ann Arbor 2019 – 2022

Frank Lab

· Machine learning for annotating RNA secondary structure and multi-task prediction of inhibitor binding

Research Assistant, University of Michigan, Ann Arbor 2018 – 2021

Walter Lab

· Riboswitch dynamics and transcription regulation using single molecule FRET and probe kinetics

Publications

†: First author, *: Corresponding author

10. **Ajmera, P.**†, Guion, D.†, Barnes, S., Sadun, A.A., Alexandrova, A.N.* (2026). Molecular Mechanism of Mitochondrial Complex I Disruption by m.14484T>C Underlying Leber Hereditary Optic Neuropathy. *Submitted, under review at PNAS*. Preprint DOI: 10.64898/2026.01.28.701874
9. Lewis, L. C., Badve, P., Vaca, I., **Ajmera, P.**, Lee, Y., Alexandrova, A.N., Meier, K., Shafaat, H.S.* (2026). A protein-based model of carbon monoxide dehydrogenase exhibits tunable covalency across cluster oxidation and ligand-bound states. *Chem. Sci., 17*(22), 11042–11061.
8. **Ajmera, P.**†, Farley, L.,†, Minnetian, N.†, King, R.P., Jayasekara, H., Chaturvedi, S., Yang, J., Alexandrova, A.N.*, Rovis, T.* (2026). Designing an Artificial Metalloenzyme for Re-based CO_2 Visible-Light Photoreduction. *Submitted, in revision at ACS Catal.*. Preprint DOI: 10.26434/chemrxiv-2025-mhw1f
7. Chaturvedi, S., Goswami, A., Qian, J., Petersen, A., **Ajmera, P.**, Alexandrova, A.N.* (2025). Distinct Electric Fields Enable Common Catalytic Function in Structurally Diverse Enzymes. *JACS, 147*(35), 32225–32237.

6. **Ajmera, P.**†, Vargas, S., Chaturvedi, S., Hennefarth, M., Alexandrova, A.N.* (2025). *PyCPET* - Computing Heterogeneous 3D Protein Electric Fields and Their Dynamics. *JCTC, 21*(8), 4299-4308.
5. Eberhart, M.*, Alexandrova, A.N.*, **Ajmera, P.**, Bím, D., Chaturvedi, S., Vargas, S., Wilson, T. (2025). Methods for Theoretical Treatment of Local Fields in Proteins and Enzymes. *Chemical Reviews, 125*(7), 3772-3813.
4. Chaturvedi, S., Vargas, S., **Ajmera, P.**, Alexandrova, A.N.* (2024). Directed Evolution Optimizes Electric Field in Protoglobin for Non-Native Catalysis. *JACS, 146*(24), 16670–16680.
3. Fuller III, J., Barnes, S., Sadun, L.A., **Ajmera, P.**, Alexandrova, A.N.*, Sadun, A.A.* (2023). Coenzyme Q10 trapping in mitochondrial Complex I underlies Leber's Hereditary Optic Neuropathy. *PNAS, 120*(39), e2304884120
2. Chauvier, A., **Ajmera, P.**, Yadav, R., Walter, N.G.* (2021). Dynamic competition between a ligand and transcription factor NusA governs riboswitch-mediated transcription regulation. *PNAS, 118*(47), e2109026118
1. Zhang, K., Abdallah, K., **Ajmera, P.**, Finos, K., Looka, A., Mekhael, J., and Frank, A.T.* (2021). CS-Annotate: A Tool for Using NMR Chemical Shifts to Annotate RNA Structure. *JCIM, 61*(4), 1545-1549.

Fellowships, Awards, & Honors

Dissertation Year Fellowship, UCLA	2026 – 2027
Jim & Barbara Tsay Excellence in Research Award, UCLA	2026
John Stauffer Excellence in Second Year Academics and Research Award, UCLA	2025
Michael E. Jung Excellence in Teaching Award, UCLA	2024
Honorable Mention, NSF GRFP	2024
NSF Quantum Science and Engineering Fellowship, UCLA	2023 – 2024
Summer Mentored Research Fellowship, UCLA	2023
University Fellowship, UCLA	2022
Exceptional Undergraduate Student Award, UMich Biophysics	2022
Distinguished Achievement Undergraduate Award, UMich Engineering Physics	2022
Ronald P. Nordgren Scholarship, UMich	2020
Charles F. Barth Jr. Prize, UMich	2020

Presentations

8. "Electrostatic Control of a Strongly Correlated $NiFe_4S_4$ Cluster", Poster at 2026 Annual Meeting of the Marcus Center of Theoretical Chemistry.
7. "Mapping dynamic 3-dimensional active-site electrostatics onto transition metal cluster properties and reactivity in enzymes", Poster at 2026 8th Southern California Theoretical Chemistry Symposium.
6. "Electrostatic Influence on the Electronic Structure of Ni,Fe Carbon Monoxide Dehydrogenase and Mimics", Poster at 2026 Metals in Biology GRC and Bioinorganic Chemistry GRS.
5. "Electrostatic Influence on the Electronic Structure of Ni,Fe Carbon Monoxide Dehydrogenase and Mimics", Talk at 2025 Southern California Bioinorganic Symposium.
4. "Evolution of Electric Fields for Catalyzing Enzyme Reactions", Poster at 2025 UCLA Glenn T. Seaborg Symposium.

3. "Electric Fields in Enzymatic Catalysis", Poster at 2024 University of California Chemical Symposium.
2. "Electric Fields in Enzymatic Catalysis", Poster at 2024 UCLA Glenn T. Seaborg Symposium.
1. "Competition between ligand binding and transcription rate modulates riboswitch-mediated regulation of transcription", Poster at 2020 Biophysical Society Conference General Poster Session and Undergraduate Poster Competition.

Teaching

TA for Chem 110B: Statistical Mechanics, UCLA	2026
TA for Chem 126A/226A: Computational Methods for Chemists, UCLA	2025
TA for Chem 113A: Introduction to Quantum Mechanics (2 quarters), UCLA	2023
TA for Biophysics 450/550: Laboratory Techniques in Biophysics (2 semesters + post-graduate), UMich	2019 – 2022
Grader for Physics 406: Statistical and Thermal Physics, UMich	2021
Curriculum Developer for Physics 401: Intermediate Mechanics, UMich	2021
TA for Knot Theory in Michigan Math & Science Scholars, UMich	2019

Research Mentorship

Giacomo Berti (Undergraduate in Biochemistry)	Feb. 2026 – Present
Bonnie Qian (Undergraduate in Chemistry)	2024 – 2025
Hiran Jayasekara (Undergraduate in Biochemistry)	2023 – 2025

Grants

NSF ACCESS: Electron Transfer in Carbon Monoxide Dehydrogenases	2024 – 2026
NSF ACCESS: Accelerating Computation and Characterization of Enzyme Electric Fields	2024 – 2025

Professional Organizations and Service

Department of Graduate Admissions, UCLA Chemistry: Reviewed and discussed graduate applications for Theory and Computation in the 2025–2026 cycle	2025 – 2026
American Chemical Society: Reviewed six manuscripts for: Journal of Physical Chemistry A & B, and ACS Catalysis	2024 – Present
Biophysical Society: University of Michigan student chapter president and representative	2020 – 2022